

- }
}
- c) Create an abstract class 'Bank' with an abstract method 'getBalance'. \$100, \$150 and \$200 are deposited in banks A, B and C respectively. 'BankA', 'BankB' and 'BankC' are subclasses of class 'Bank', each having a method named 'getBalance'. Call this method by creating an object of each of the three classes. 6
- Q.3 a) Write a Java program that does the following: 6
- Declare a string object named s containing the string "This is extraordinary!!".
 - Print the entire string.
 - Use the length() method to print the length of the string.
 - Use the charAt() method to print the first character in the string.
 - Use the charAt() and the length () methods to print the last character in the string.
 - Use the indexOf() and the substring() methods to print the first word in the string.
- b) Write a Java program that inputs three names and then prints them in their increasing alphabetical order. Use the String class method compareTo. 6
- c) Differentiate PushBackInputStream and ByteArrayInputStream classes. Write a program using ByteArrayInputStream class to read from byte array and convert desired characters into uppercase. 6
- Q.4 a) Explain Java thread model by comparing multithreading environment with Single thread System. Explain different type of priorities associated with thread using simple programming example. 6
- b) What is Producer/Consumer relationship problem? Write a program to implement a queue of objects. Use this queue to implement producer and consumer classes. Use wait() and notify() methods to solve this problem. 6
- c) What is meant by Exception handling approach? Write a program by creating subclass of IOException to handle an Exception in input given by user. 6
- Q.5 a) What are the different types of JDBC Statements and what are JDBC driver types illustrate via suitable code? 6
- b) What is the difference between execute, executeQuery, executeUpdate? 6
What is the limitation of PreparedStatement and how to overcome it?
- c) Write a Java Program to read a string in the form of command line arguments and print them till 'stop' comes 6

***** End of the Question Paper*****